

File permissions in Linux

Project description

Linux is mainly a CLI OS. This means that the optimal way to interact with our computer is through commands written on the bash shell. As a security analyst, you must perform several tasks using this tool. In this case, we are reviewing how to manage permissions employing commands such as `chmod`, `ls -l`, `ls -a` and `ls -la`.

To complete the task given, we must follow these steps. First I will check both user and group permissions for all the files in the `projects` directory. Then I will check if any file has incorrect permissions and modify them as necessary. Finally, I will check the permissions of the `/home/researcher2/drafts` directory and modify them to remove unauthorized access.

Check file and directory details

The first line demonstrates the command I have entered to display all the permissions of both files and directories or subdirectories stored in the `projects` directory.

To obtain this result, I have used the `ls` command with the `-la` option. Let's understand what are the specific uses of each option. The `-l` option will display all the permissions for files and directories, while the option `-a` displays hidden files, identified with `(.)` at the beginning. Combined, they display permissions for files and directories, including hidden files.

```
researcher2@57813a649b08:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 02:32 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 02:55 ..
-rw--w---- 1 researcher2 research_team  46 Aug 31 02:32 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 31 02:32 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 31 02:32 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 31 02:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_t.txt
researcher2@57813a649b08:~/projects$
```

Describe the permissions string

In Linux, file permissions are represented with a 10-character string.

```
researcher2@57813a649b08:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 02:32 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 02:55 ..
-rw--w---- 1 researcher2 research_team  46 Aug 31 02:32 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 31 02:32 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 31 02:32 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 31 02:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_t.txt
researcher2@57813a649b08:~/projects$
```

To understand how it is composed, we must get in contact with two labels.

The first one includes the **types of permissions** formed by, **read** files and directories can be read, **write** files and directories can be modified and created respectively and **execute**, files can be executed and directories can be opened by accessing their files.

The second includes the **types of owners** formed by the **user** owner of the file, a **group** of users and **others** that can be considered all the other users on the system.

Once we get in touch with these concepts, we can break down the 10-character string

The first character is used to indicate if it is a file or a directory, using a **-** for files like the yellow one and a **d** for directories like the red one.

Then each type of the 3 owners that we have explained has assigned 3 characters for each type of permission: read represented with an **r**, write represented with a **w** and execute represented with an **x**. Also, if any of the permissions are not assigned, it will be an **-** instead. Consequently, between 2nd-4th characters we can identify the user permissions, between 5th-7th characters the group permissions and between 8th-10th characters the other permissions.

Change file permissions

The organization determined that other shouldn't have write access to any of their files. To achieve this, I have used the `chmod` command used to manage authorization. As can be seen on the first line, the input command structure is `chmod (command) o-w (first argument)` and `project_k.txt` (second argument).

To break down the first argument, we have to understand that we should use `o` for other owner types, `g` for group and `u` for user. Also, to add or remove permissions, we should use plus (+) or equal (=) to add and minus (-) to remove.

```
researcher2@57813a649b08:~/projects$ chmod o-w project_k.txt
researcher2@57813a649b08:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 02:32 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 02:55 ..
-rw--w---- 1 researcher2 research_team  46 Aug 31 02:32 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 31 02:32 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 31 02:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_t.txt
researcher2@57813a649b08:~/projects$
```

Change file permissions on a hidden file

The organization has decided that the hidden file `.project_x.txt` should only be read by user and group. To achieve this task I have used this command:

```
researcher2@57813a649b08:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@57813a649b08:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 02:32 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 02:55 ..
-r--r----- 1 researcher2 research_team  46 Aug 31 02:32 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 31 02:32 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 31 02:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 02:32 project_t.txt
researcher2@57813a649b08:~/projects$
```

In this case, I modified the group permissions using ``g+r`` to grant read access to the group, while removing the write permission via ``g-w`` and, finally, revoking the user's write permissions with ``u-w``.

Change directory permissions

The organization has decided that only the user `researcher2` should be able to access the `drafts` directory and its contents. To complete this task I have used again the `chmod` command but in this case for a directory.

```
researcher2@b8ec57f7e4f2:~/projects$ chmod g-x drafts
researcher2@b8ec57f7e4f2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 03:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 31 04:15 ..
-rw--w---- 1 researcher2 research_team  46 Aug 31 03:55 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Aug 31 03:55 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 31 03:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 31 03:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 03:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 31 03:55 project_t.txt
researcher2@b8ec57f7e4f2:~/projects$
```

Summary

My tasks were completed. Using the right commands I have checked the permissions display of my organization with `ls -la` command. Then with the instructions given I have modified some permissions including files, hidden files and directories using `chmod`.